Economic Analysis with Matrices (ECO2048)

Final Project

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Semester 1 – 2023-2024

Instructions

- **Deadline: January 3th, 2024 by 4pm**.
- Please upload this as a .zip file in the submission folder ECO2048 - Final Assignment on Surrey Learn. **Important:** do not forget to label the submission folder with your group number.
- The .zip file should include:
  - MATLAB codes (properly referencing the relevant question). Please also state which MATLAB version you are using (e.g. 2023b).
  - A report with fully worked out solutions.
  - A self-assessed individual contribution form.
- **Legend:** [**C**]: answer requires MATLAB code; [**R**]: answer requires explanation in the report.

Notes

- The quality and readability of your report and MATLAB codes will be taken into consideration in the final mark. Please ensure that your report is tidy, well-formatted and organised.
- Unfortunately, I will not be able to accept scanned solutions and/or handwritten copies. You have to write the equations/solutions in digital format. The same applies to all tables and graphs.

1 The BLUE Project (50% of total marks)

1. Download 5 variables of your choice from the FRED database with MATLAB. Using your own words, give a brief description of each variable and **explain why they are relevant from an economic point of view**. [**C**]

   A word on notation: Let n be the number of observations. We use $\mathbf{x}_1$ to denote the first variable ($n \times 1$), $\mathbf{x}_2$ the second ($n \times 1$), and so on until $\mathbf{x}_4$. We use $\mathbf{y}$ to denote the fifth variable ($n \times 1$). We stack all the $\mathbf{x}$'s in the $n \times 4$ matrix $\mathbf{X} = [\mathbf{x}_1\ \mathbf{x}_2\ \mathbf{x}_3\ \mathbf{x}_4]$.

2. Describe the units of measurement for each variable, and make any suitable transformations to the original data if needed. If you do so, please motivate your choices. [**C**] [**R**]

   For the avoidance of doubt: if you have transformed the variables following task 2, please use the transformed variables to answer all remaining questions in this section.

3. How do you expect $\mathbf{y}$ and $\mathbf{x}_1$ (the first variable) to be related? Show (a) a scatterplot of $\mathbf{y}$ on $\mathbf{x}_1$ and (b) a plot of both $\mathbf{y}$ and $\mathbf{x}_1$ over time. Is the association between $\mathbf{y}$ and $\mathbf{x}_1$ as you expected before seeing the data? Comment. [**C**] [**R**]

4. Report the following descriptive statistics for both $\mathbf{y}$ and $\mathbf{X}$: number of observations, average, standard deviation, median, minimum, maximum, and the correlations involving the 5 variables. [**C**] [**R**]

5. Run Ordinary Least Squares (OLS) regression of $\mathbf{y}$ on $\mathbf{X}$ and an intercept using a built-in function. Report the estimated parameters (there should be 5, one for the intercept and 4 for the $\mathbf{x}$'s variables). Implement the formula for the estimated parameters with explicit matrix computations, and compare your estimates to the ones obtained with the built-in function. [**C**] [**R**]

6. Run OLS of $\mathbf{y}$ on $\mathbf{x}_1$ and an intercept. Compare the coefficient on $\mathbf{x}_1$ with the one obtained in the previous question, and give an intuitive explanation for the difference (if any). [**C**] [**R**]

7. Plot the regression line (task 6) superimposed on the scatterplot (task 3). Note that in practice plotting the regression line involves plotting an $n \times 1$ vector $\hat{\mathbf{y}} = \hat{\beta}_0\mathbf{1} + \hat{\beta}_1\mathbf{x}_1$, where $\mathbf{1}$ is the $n \times 1$ vector of ones. Implement the formula in MATLAB with explicit matrix calculations. [**C**] [**R**]

8. Compute the eigendecomposition of the sample covariance matrix of $\mathbf{X}$. Report the eigenvalues in decreasing order. [**C**] [**R**]

9. Following task 8, let $\mathbf{v}$ be the eigenvector corresponding to the largest eigenvalue. Compute

$$\mathbf{z} = (\mathbf{X} - \mathbf{1}\mathbf{1}^\top\mathbf{X}/n)\mathbf{v},$$

   where $\mathbf{1}$ is the $n \times 1$ vector of ones. Can you give an interpretation of $\mathbf{z}$? Show a scatterplot of $\mathbf{y}$ on $\mathbf{z}$ with a regression line superimposed.

10. Let $\mathbf{M} = [\mathbf{1}\ \mathbf{X}]$ denote the $n \times 5$ matrix obtained by stacking $\mathbf{1}$ and $\mathbf{X}$ horizontally. Compute and report the following:[1]

$$C = \frac{\lambda_{\min}(\mathbf{M}^\top\mathbf{M})}{\lambda_{\max}(\mathbf{M}^\top\mathbf{M})},$$

    where $\lambda_{\min}(\mathbf{M}^\top\mathbf{M})$ is the minimum eigenvalue of $\mathbf{M}^\top\mathbf{M}$, and $\lambda_{\max}(\mathbf{M}^\top\mathbf{M})$ is the maximum eigenvalue of $\mathbf{M}^\top\mathbf{M}$. [**C**] [**R**]

[1] The number C is known as the **reciprocal condition number**. If it is close to 2e-16 (this is 2 times 0.000...1 where the 1 is in the 16th decimal place), the $\mathbf{M}^\top\mathbf{M}$ matrix is **ill-conditioned**. Ill-conditioned regressors have the potential problem that the estimated parameters may be inaccurate.

2 The Buffett Project (50% of total marks)

1. Download the data for the universe of stocks listed in the file `tickers.txt`, from December 31st 2021 to December 31st 2022 (note: there were 252 trading days in this period).[2] [**C**]

2. If a stock has less than 252 available trading days, discard it. Out of the remaining assets, pick 100 using a random generator. For each selected stock, keep only the Adjusted Close price information. [**C**]

3. Compute daily returns following the formula $100\left(\frac{P_t}{P_{t-1}} - 1\right)$, where P_t is the Adjusted Close price on day t. Do this for each selected stock, and store the results in a 251×100 matrix. [**C**]

4. Compute the sample covariance matrix $\mathbf{S}$ of daily returns. This should be a 100×100 matrix. [**C**]

5. Choose 11 evenly-spaced values of α between 0 and 1 (included), and compute the following for each value of α:

$$\mathbf{S}_\alpha = \alpha\mathbf{S} + (1-\alpha)\hat{\sigma}^2\mathbf{I},$$

   where $\hat{\sigma}^2$ is the sample variance of all assets in your portfolio (see the documentation for the `var` function). Explain how you may store the 11 matrices in a convenient manner. [**C**] [**R**]

6. Compute the minimum variance portfolio weights using the sample covariance matrix $\mathbf{S}$. Report the top 5 and bottom 5 stocks based on their portfolio weights. [**C**] [**R**]

7. Repeat task 6 but now use the regularized estimator $\mathbf{S}_\alpha$ for each value of α. [**C**] [**R**]

8. Explain the results obtained in the cases $\alpha = 0$ and $\alpha = 1$. How do the weights behave as we move α from 0 to 1? [**R**]

9. Discuss how to evaluate and compare the performance of your portfolios. [**R**]

10. Download the data for 2023 and use the portfolio weights based on year 2022 to evaluate the performance of your portfolio in 2023 for each value of α. What do you conclude? [**C**] [**R**]

[2] You may find the function `hist_stock_data.m` useful for this task.